# Control of particulate matter

# What is particulate matter?

Particulates are defined as any dispersed solid matter,, in which the individual aggregates are larger than a single small molecule (about 0.002 µm in diameter) but smaller than about 500 µm.

# Classification

Classified and discussed according to their physical, chemical, or biological characteristics.

- Physical char: size, mode of formation, settling properties, and optical qualities.
- Chemical char: organic and inorganic composition.
- Biological char: bacteria, viruses, pollens, spores etc.

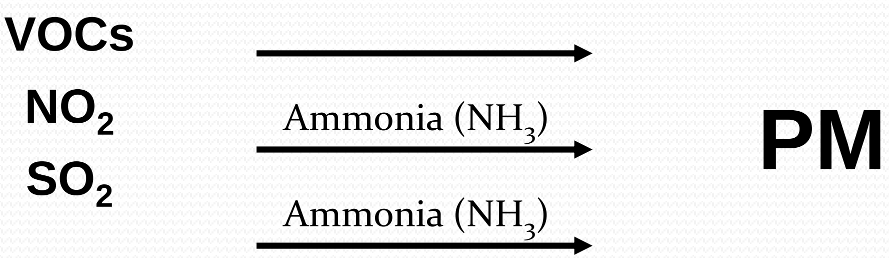
# How is PM regulated?

PM is one of the six **EPA "criteria pollutants"** that have been determined to be harmful to public health and the environment. (The other five are ozone, sulfur dioxide, nitrogen dioxide, carbon monoxide, and lead.)

EPA is required under the Clean Air Act to set **national ambient air quality standards (NAAQS)** to protect public health from exposure to these pollutants. Areas that exceed the NAAQS are designated as **nonattainment**, and must institute air pollution control programs to reduce air pollution to levels that meet the NAAQS.

# Where does PM originate?

Sources may emit PM directly into the environment or emit **precursors** such as **sulfur dioxide ($SO_2$), nitrogen dioxide ($NO_2$), and volatile organic compounds (VOCs),** which are transformed through atmospheric chemistry to form PM.



# Sources of PM and PM Precursors

**Mobile Sources**
(vehicles)
VOCs, $NO_2$, PM

**Stationary Sources**
(power plants, factories)
$NO_2$, $SO_2$, PM

**Area Sources**
(drycleaners, gas stations)
VOCs

**Natural Sources**
(forest fires, volcanoes)
PM

# What adverse effects have been linked to PM?

- Premature death
- Lung cancer
- Development of chronic lung disease
- Heart attacks
- Decreased lung function
- Pre-term birth
- Low birth weight
- Damage materials by soiling clothing and textiles ,corroding metals, eroding building surfaces, and discoloring and destroying painted surfaces.
- Damage plant tissues
- Ill effects on animals who eats plants coated with particulates containing fluorides, arsenic or lead.
- Reduction in visibility

# How does PM cause health effects?

Several theories have been advanced as to the mechanism of action. It is likely that more than one mechanism is involved in causing PM-related health effects. Theories include the following:

1. PM leads to lung **irritation** which leads to increase permeability in lung tissue;
2. PM increases **susceptibility to viral and bacterial pathogens** leading to pneumonia in vulnerable persons who are unable to clear these infections;
3. PM **aggravates the severity of chronic lung diseases** causing rapid loss of airway function;
4. PM causes **inflammation** of lung tissue, resulting in the release of chemicals that impact heart function;
5. PM causes **changes in blood chemistry** that results in clots that can cause heart attacks.

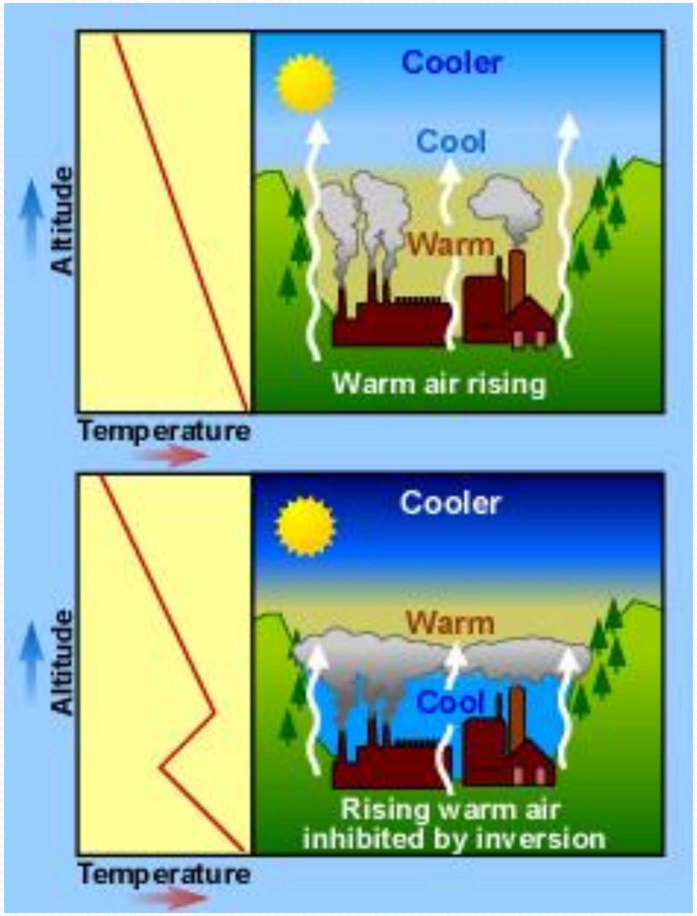
# Increasing Evidence of Cardiovascular Effects

Until the mid 1990s, most research focused on the association of PM exposure with respiratory disease. Since then, there has been growing evidence of **cardiovascular health effects** from PM.

# Particulate Matter: Size Matters

Size is important to the behavior of PM in the atmosphere and human body and determines the entry and absorption potential for particles in the lungs. Particles larger than 10 µm are trapped in the nose and throat and never reach the lungs. Therefore, particles 10 µm in diameter or less are of most concern for their effects on human health. Particles between 5 and 10 µm are removed by physical processes in the throat. Particles smaller than 5 µm reach the bronchial tubes, while particles 2.5 µm in diameter or smaller are breathed into the deepest portions of the lungs.

# The Role of Inversions




Source: http://www.epa.gov/apti/course422/ ce1.html

**An inversion** is an extremely stable layer of the atmosphere that forms over areas.

Temperature inversions trap pollutants close to the ground. These inversions involve layers of hot air sitting above cooler air near ground level. When particles accumulate in the air layer, they are unable to rise into the atmosphere where winds will disperse them.

# Major Episodes of Severe Air Pollution due to Inversions

1930: Meuse River Valley, Belgium

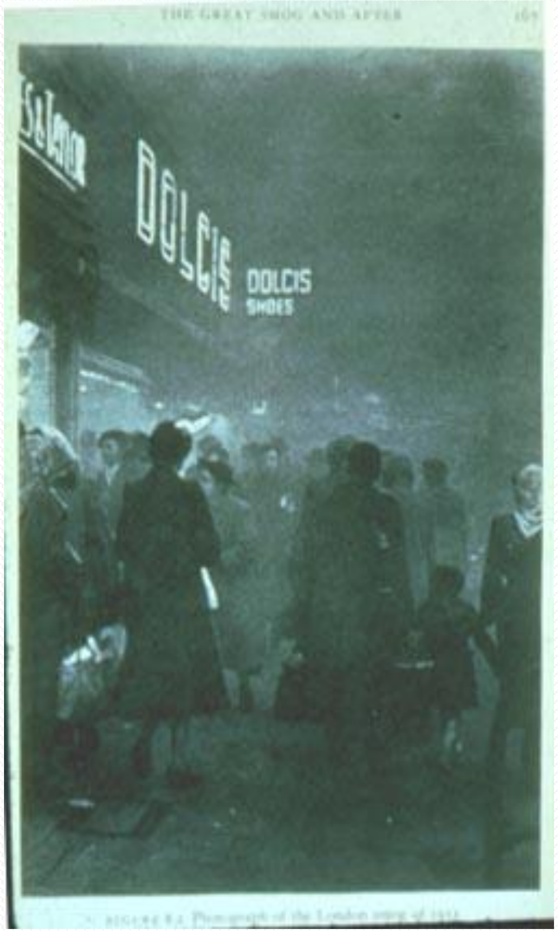
- An inversion led to a high concentration of pollutants during a period of cold, damp weather
- Main sources: zinc smelter, sulfuric acid factory, glass manufacturers
- 60 deaths recorded

1948: Donora, Pennsylvania

- Similar inversion to Meuse River Valley
- Main sources: iron and steel factories, zinc smelting, and an acid plant
- 20 deaths observed

1952: London

- Killer fog (right)
- Primary source: domestic coal burning
- 4,500 excess deaths recorded during week- long period in December




**The Great London Smog (1952)**

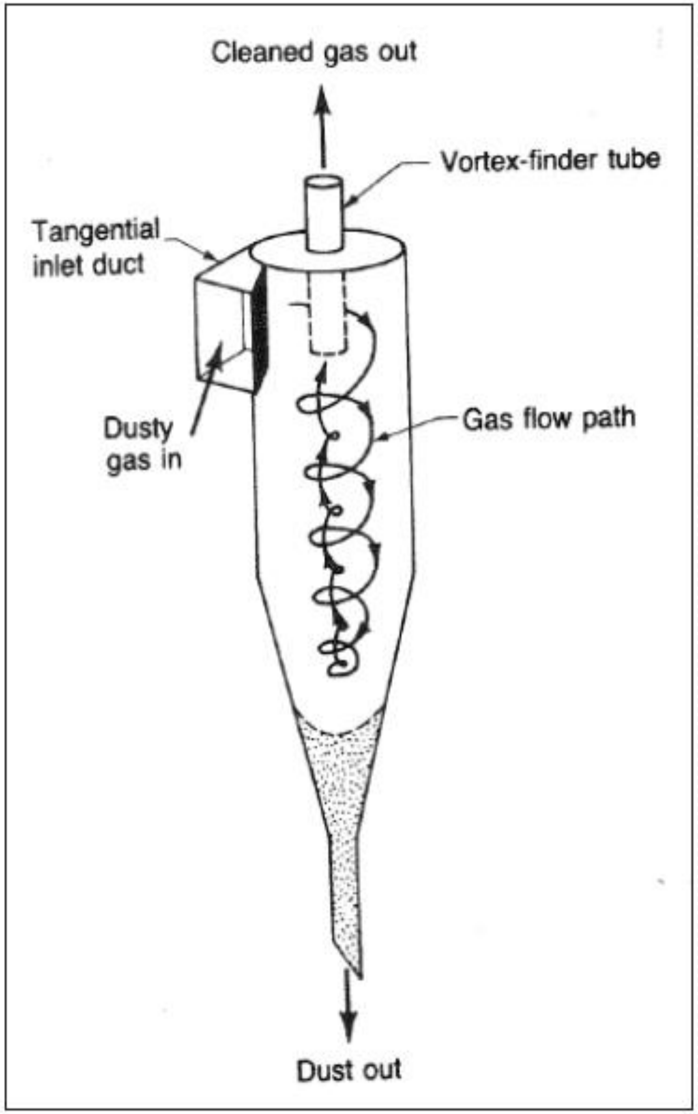
# Control of particulate pollutants

The most commonly used devices for controlling particulate emissions include:

- electrostatic precipitators (wet and dry types)
- fabric filters (also called bag houses),
- wet scrubbers including spray towers, wet cyclone scrubbers and venturi-scrubbers
- gravitational settling chambers
- Centrifugal collectors including cyclone collectors and dynamic precipitators.

Source: S.K.Garg, sewage disposal and air pollution engg., vol 2, pg. 811

# Cyclone collector




A cyclone collector is a specially designed closed chamber in which the velocity of the inlet gas is transformed into spinning vortex, and the particles from the gas are thrown out under the centrifugal force. The particles thrown out on the walls of the chamber, slide down to the hopper, and, are thus removed.

- minimum particle size that can be removed: 5-25μm
- efficiency:50-90%
- merits: 1. relatively inexpensive, simple to design and maintain.
  2. requires less floor area.
  3. low to moderate pressure loss.
  4. can handle large volume of gases at temperature up to 90 °c.
- Demerits: 1. collection efficiency is low for smaller particles.
  2. sensitive to variable dust loadings and flow rates.

Source: S.K.Garg, sewage disposal and air pollution engg., vol 2, pg. 814

# Gravitational settling chamber

principle of working:

The emitted smokes, when made to pass through a settling chamber, drop some of their larger sized particles in the chamber, under stoke’s law.

- minimum particle size that can be removed: >50µm
- efficiency: <50%
- merits: 1. simple to design and maintain
  2. low pressure loss
- demerits: 1. requires larger space
  2. collection efficiency is also low
  3. only larger sized particles are separated out

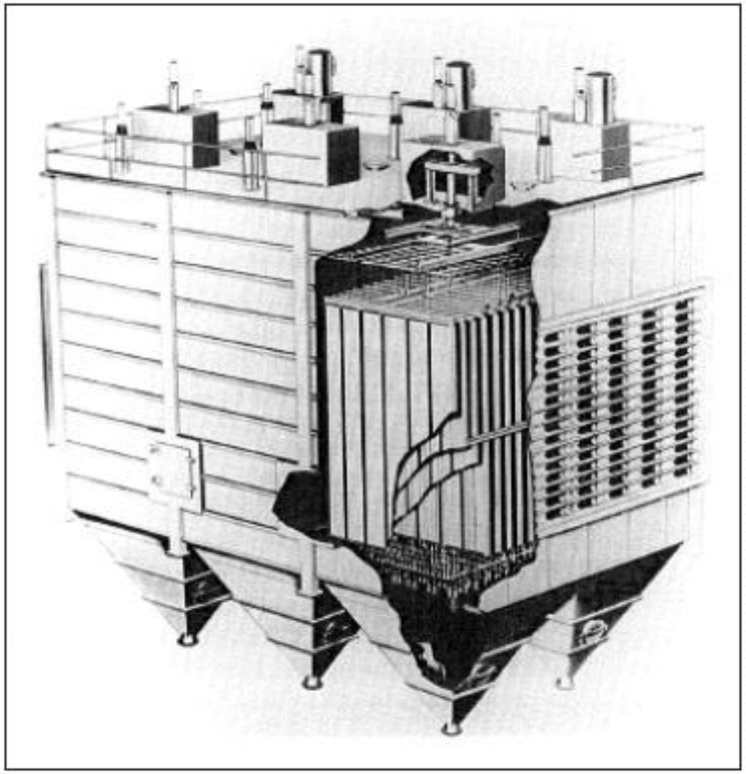
Source: S.K.Garg, sewage disposal and air pollution engg., vol 2, pg. 812

# Wet Scrubbers

- Includes spray tower, wet cyclonic scrubber and venturi-scrubber
- In these devices, the flue gas is made to push up against a down falling water current. The particulate matter mixes up with the water droplets and, thus, falls down and gets removed.

- merits: 1. The spray tower and venturi scrubber can be made to remove gaseous pollutants also, along with removing particulate matter
  2. hot gases can be cooled down.
  3. corrosive gases can be recovered and neutralized.
- demerits: 1. maintenance cost is high, when corrosive materials are collected.
  2. wet outlet gases cannot rise high from the stack

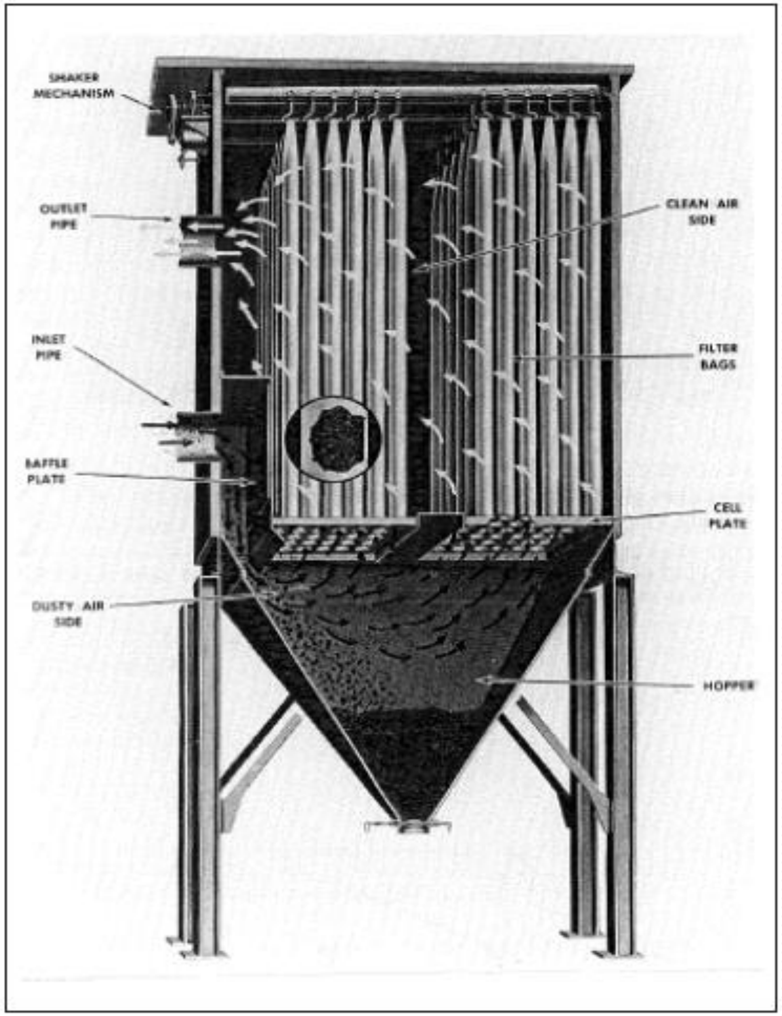
# Electrostatic precipitators



In electrostatic precipitators, the flue gas is made to pass through a highly ionized zone, where the particles get electrically charged and are separated out from the gas, with the help of electrostatic forces in the powerful electric field.

- minimum particle size that can be removed: >1µm
- efficiency: 95-99%
- merits: 1. particles may be collected wet or dry
  2. even small particles can be removed.
  3. they contain a few moving parts.
  4. can operate at high temperature up to 300-450°c
- demerits: 1. higher initial costs.
  2. sensitive to variable dust loadings and flow rates.
  3. collection efficiency reduces with time.

# Fabric filters(bag houses)




In such a system, the flue gas is allowed to pass through a woven or felted fabric, which filters out the particulate matter and allows the gas to pass. Small particles are retained on the fabric, initially through interception and electrostatic attraction; and later on, when a dust mat is formed, the fabric starts collecting particles more efficiently.

- minimum particle size that can be removed: <1μm
- <u>efficiency</u>: >99%
- <u>merits</u>: 1. can remove very small particles in dry state.
  2. give high efficiency.
  3. performance decrease becomes visible, gives prewarning.
- <u>demerits</u>: 1. fabric is liable to chemical attack.
  2. flue gases must be dry, otherwise, there is a risk of condensation inside the filter, which can cause clogging.

Source: S.K.Garg, sewage disposal and air pollution engg., vol 2, pg. 822

## References: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# Thank you!